
**CS301 MID TERM PAST PAPERS
(SOLVED MCQS)
(VISIT VU RANK FOR MORE)**

MTH301 Calculus II Marks: 1 (Budgeted Time 1 Min)

Question No : 1 of 26

Every real number corresponds to ----- on the co-ordinate line.

Answer (Please select your correct option)

☒ Infinite number of points

☐ Two points (one positive and one negative)

☐ A unique point

☐ None of these

MTH301 Calculus II Marks: 1 (Budgeted Time 1 Min)

Question No : 2 of 26

There is one-to-one correspondence between the set of points on co-ordinate line and -----

Answer (Please select your correct option)

☐ Set of real numbers

☐ Set of integers

☒ Set of natural numbers

☐ Set of rational numbers

Which of the following is associated to each point of three dimensional space?

☐ A real number

☐ An ordered pair

☐ An ordered triple

☒ A natural number

All axes are positive in ----- octant.

☒ First

☐ Second

☐ Fourth

☐ Eighth

The spherical co-ordinates of a point are $\left(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2}\right)$. What are its cylindrical co-ordinates?

☐ $\left(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0\right)$

☐ $\left(\sqrt{3} \cos \frac{\pi}{3}, \sqrt{3} \sin \frac{\pi}{3}, 0\right)$

☒ $\left(\sqrt{3} \sin \frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3} \cos \frac{\pi}{3}\right)$

☐ $\left(\sqrt{3}, \frac{\pi}{3}, 0\right)$

Suppose $f(x, y) = xy - 2y^2$ where $x = 3t + 1$ and $y = 2t$. Which one of the following is true?

☒ $\frac{df}{dt} = -4t + 2$

☐ $\frac{df}{dt} = -16t - t$

☐ $\frac{df}{dt} = 18t + 2$

☐ $\frac{df}{dt} = -10t^2 + 8t + 1$

Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$, $z = t(r, s)$ then by chain rule

$$\frac{\partial w}{\partial r} =$$

Answer (Please select your correct option)

☐ $\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r}$

☐ $\frac{\partial w}{\partial r} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial r} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial r} \frac{\partial z}{\partial r}$

☒ $\frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s}$

☐ $\frac{\partial w}{\partial r} \frac{\partial x}{\partial x} + \frac{\partial w}{\partial r} \frac{\partial y}{\partial y} + \frac{\partial w}{\partial r} \frac{\partial z}{\partial z}$

Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

Answer (Please select your correct option)

☐ 4.5

☒ 6.2

☐ 5.1

☐ 4.2

MTH301 Calculus II

Question No : 9 of 26

Marks: 1 (Budgeted Time 1 Min)

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} 0 & \text{If } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{Otherwise} \end{cases}$$

Answer (Please select your correct option)

- ☐ $f(x, y)$ is continuous at origin
- ☒ $f(0, 0)$ is not defined
- ☐ $f(0, 0)$ is defined but $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist
- ☐ $f(0, 0)$ is defined and $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists but these two numbers are not equal

MTH301 Calculus II

Question No : 10 of 26

Marks: 1 (Budgeted Time 1 Min)

Is the function $f(x, y)$ continuous at origin? If not, why?

$$f(x, y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases}$$

Answer (Please select your correct option)

- ☐ $f(x, y)$ is continuous at origin
- ☒ $f(0, 0)$ is not defined
- ☐ $f(0, 0)$ is defined but $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist
- ☐ $f(0, 0)$ is defined and $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists but these two numbers are not equal

Let R be a closed region in two dimensional space. What does the double integral over R calculates?

Answer (Please select your correct option)

- ☐ Area of R .
- ☒ Radius of inscribed circle in R .
- ☐ Distance between two endpoints of R .
- ☐ None of these

Which of the following formula can be used to find the Volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

Answer (Please select your correct option)

- ☐ $|\vec{a} \times (\vec{b} \times \vec{c})|$
- ☒ $|\vec{a} \cdot (\vec{b} \times \vec{c})|$
- ☐ $|\vec{a} \cdot (\vec{b} \times \vec{c})|$
- ☐ $|\vec{a} \times (\vec{b} \cdot \vec{c})|$

Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are -----

Answer (Please select your correct option)

☒ Parallel

☐ Perpendicular

☐ In opposite direction

By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R , then $f(x, y)$ has both ----- on R .

Answer (Please select your correct option)

☒ Absolute maximum and absolute minimum value

☐ Relative maximum and relative minimum value

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has -----

Answer (Please select your correct option)

☒ Relative maximum at (x_0, y_0)

☐ Relative minimum at (x_0, y_0)

☐ Saddle point at (x_0, y_0)

☐ No conclusion can be drawn.

Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy} \text{ and } f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$

If $D = 0$ then -----

Answer (Please select your correct option)

☒ f has relative maximum at (x_0, y_0)

☐ f has relative minimum at (x_0, y_0)

☐ f has saddle point at (x_0, y_0)

☐ No conclusion can be drawn.

If $R = R_1 \cup R_2$, where R_1 and R_2 are no overlapping regions then

$$\iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA =$$

Answer (Please select your correct option)

☐ $\iint_R f(x, y) dA$

☐ $\iint_{R_1} f(x, y) dA \cup \iint_{R_2} f(x, y) dA$

☐ $\iint_R f(x, y) dV$

☒ $\iint_{R_1} f(x, y) dA \cup \iint_{R_2} f(x, y) dA$

If $R = \{(x, y) / 0 \leq x \leq 2 \text{ and } 1 \leq y \leq 4\}$, then

$$\iint_R (6x^2 + 4xy^3) dA =$$

Answer (Please select your correct option)

☐ $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dy dx$

☒ $\int_0^2 \int_1^4 (6x^2 + 4xy^3) dx dy$

☐ $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dx dy$

☐ $\int_2^4 \int_0^1 (6x^2 + 4xy^3) dx dy$

MTH301 Calculus II

Question No : 19 of 26

Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) / 2 \leq x \leq 4 \text{ and } 0 \leq y \leq 1\}$, then

$$\iint_R (4xe^{2y}) dA =$$

Answer (Please select your correct option)

☐ $\int_0^1 \int_2^4 (4xe^{2y}) dy dx$

☒ $\int_0^1 \int_2^4 (4xe^{2y}) dx dy$

☐ $\int_1^4 \int_0^2 (4xe^{2y}) dx dy$

☐ $\int_1^4 \int_0^2 (4xe^{2y}) dy dx$

MTH301 Calculus II

Question No : 20 of 26

Marks: 1 (Budgeted Time 1 Min)

If $R = \{(x, y) / 0 \leq x \leq 4 \text{ and } 0 \leq y \leq 9\}$, then

$$\iint_R (3x - 4x\sqrt{xy}) dA =$$

Answer (Please select your correct option)

☐ $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy}) dy dx$

☐ $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy}) dx dy$

☐ $\int_0^9 \int_0^4 (3x - 4x\sqrt{xy}) dx dy$

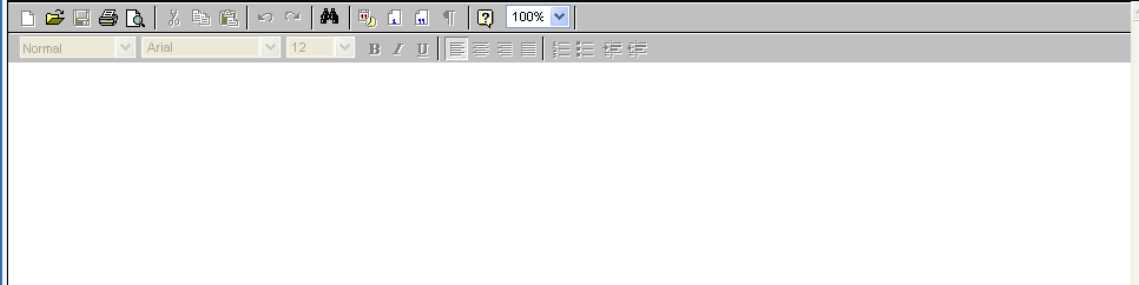
☒ $\int_0^4 \int_0^9 (3x - 4x\sqrt{xy}) dy dx$

MTH301 Calculus II

Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

Suppose that the surface $f(x, y, z)$ has continuous partial derivatives at the point (a, b, c) . Write down the equation of tangent plane at this point.

Answer (Please [click here](#) to Add Answer)

MTH301 Calculus II

Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Evaluate the following double integral.

$$\iint (12xy^2 - 8x^3) \, dy \, dx$$

MTH301 Calculus II

Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Evaluate the following double integral.

$$\iint (3 + 2x - 3y^2) \, dx \, dy$$

Answer (Please [click here](#) to Edit Answer)

MTH301 Calculus II

Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Let $f(x, y, z) = xy^2e^z$

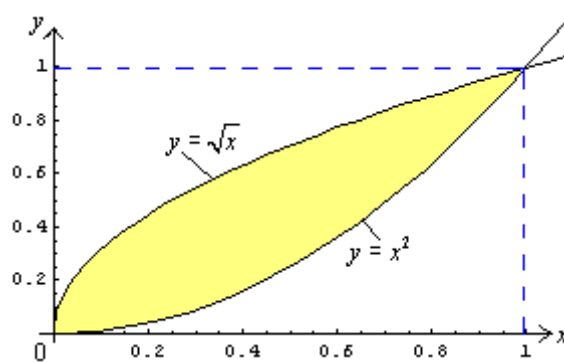
Find the gradient of f .

Answer (Please [click here](#) to Add Answer)

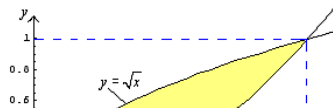
Find, Equation of Tangent plane to the surface $f(x, y, z) = x^2 + y^2 + z - 9$ at the point $(1, 2, 4)$

Answer (Please [click here](#) to Add Answer)

Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



Use double integral in rectangular co-ordinates to compute area of the region bounded by the curves $y = x^2$ and $y = \sqrt{x}$.



Answer (Please [click here](#) to Add Answer)